

What Cross-Modal Connectome Prediction Does and Does Not Buy

Reconstruction, baselines, and downstream utility (Glasser; 4S456 in backup)

Adel Sahuc

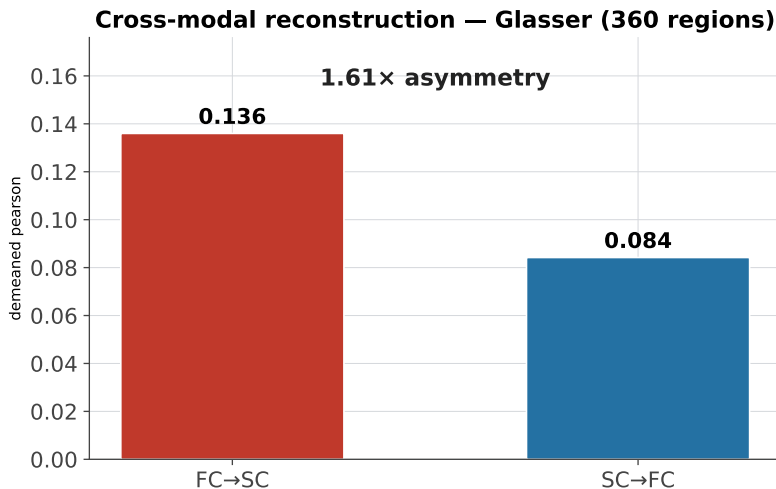
The question

- ▶ If FC predicts SC (or vice versa), can the *predicted* connectome stand in for the missing one?
- ▶ Two sub-questions: (1) can we *reconstruct* it? (2) is the reconstruction *useful*?
- ▶ All numbers: HCP-YA, 10 frozen family-aware splits; reconstruction in demeaned pearson, downstream in BayesianRidge lift over a cheap bv+demo baseline.

1 · Cross-modal prediction is directional [PCA→PLS]

direction	demeaned r
FC→SC	0.136
SC→FC	0.084
ratio	1.61 ×

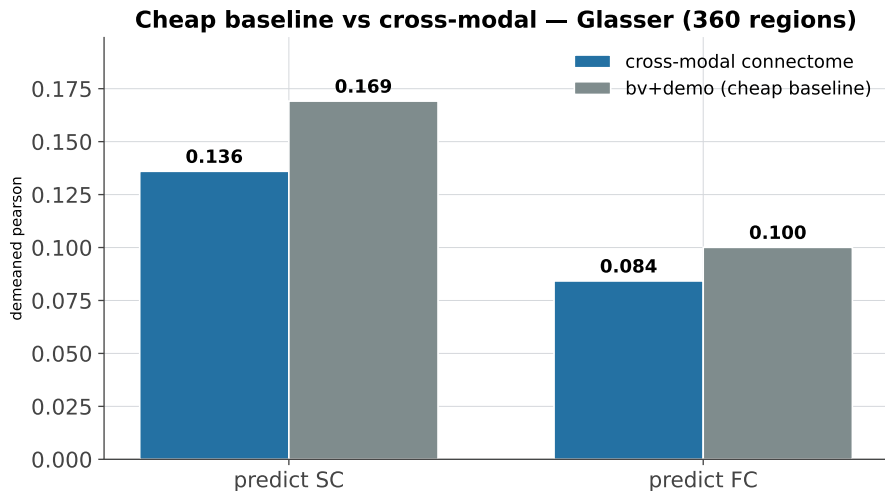
1 · Cross-modal prediction is directional [PCA→PLS]



2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS]

target	cross-modal	bv+demo
→ SC	0.136	0.169
→ FC	0.084	0.100

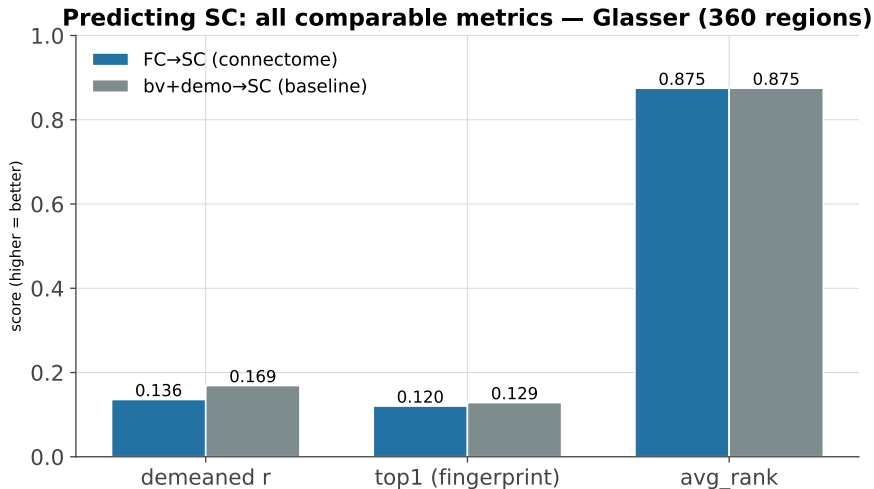
2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS]



3a · Predicting SC: every metric [PCA→PLS]

input	demeaned r	raw r	top1	avg rank	mse	r^2
FC→SC	0.136	0.913	0.120	0.875	0.006	-0.021
bv+demo→SC	0.169	0.914	0.129	0.875	0.005	0.008

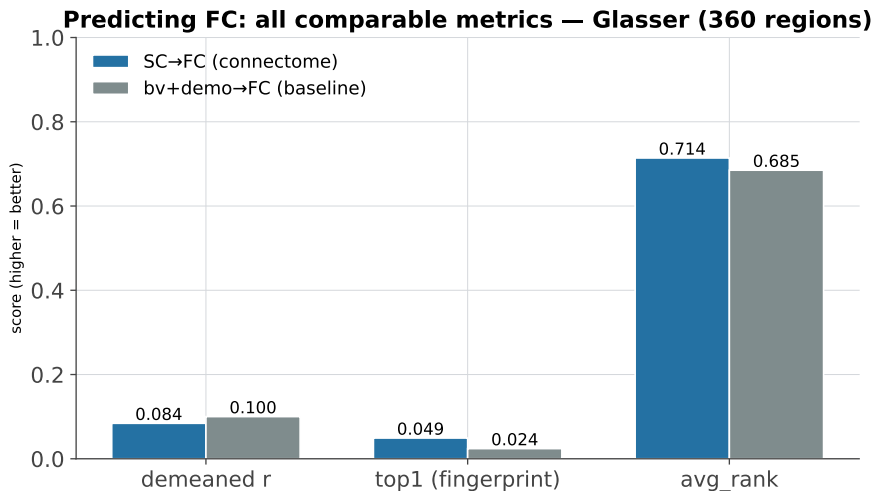
3a · Predicting SC: comparable metrics [PCA→PLS]



3b · Predicting FC: every metric [PCA→PLS]

input	demeaned r	raw r	top1	avg rank	mse	r^2
SC→FC	0.084	0.828	0.049	0.714	0.014	-0.056
bv+demo→FC	0.100	0.834	0.024	0.685	0.014	-0.019

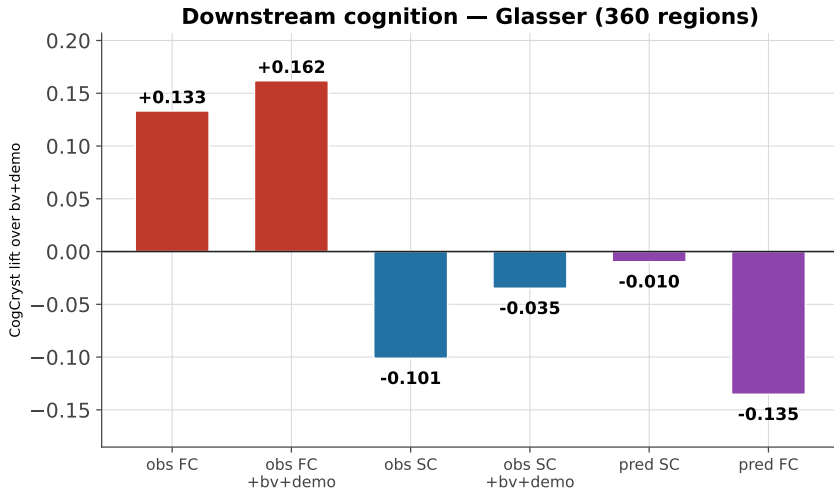
3b · Predicting FC: comparable metrics [PCA→PLS]



4 · Downstream cognition: only observed FC helps [BayesianRidge]

input	CogCryst r	lift vs bv+demo
observed FC	0.487	+0.133
observed FC + bv+demo	0.516	+0.162
observed SC	0.253	-0.101
observed SC + bv+demo	0.319	-0.035
predicted SC	0.344	-0.010
predicted FC	0.219	-0.135

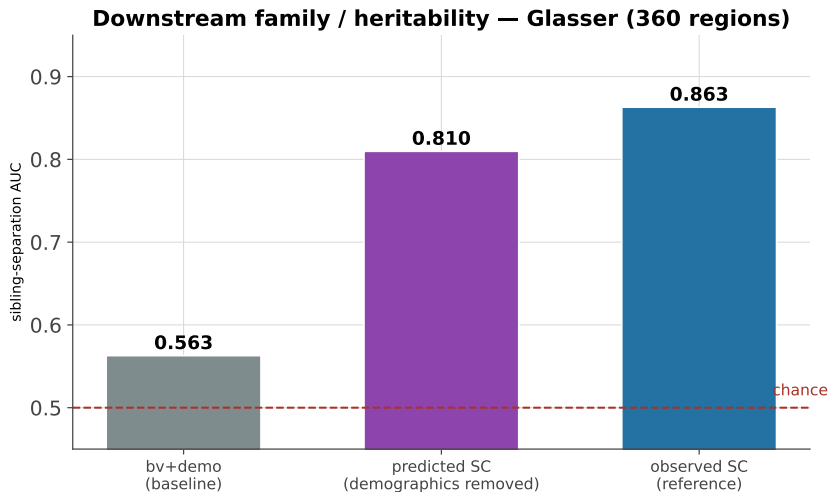
4 · Downstream cognition: only observed FC helps [BayesianRidge]



5 · Downstream family: the predicted connectome carries it [PCA→PLS]

predictor	sibling AUC
bv+demo (baseline)	0.563
predicted SC (demographics removed)	0.810
observed SC (reference)	0.863

5 · Downstream family: the predicted connectome carries it [PCA→PLS]



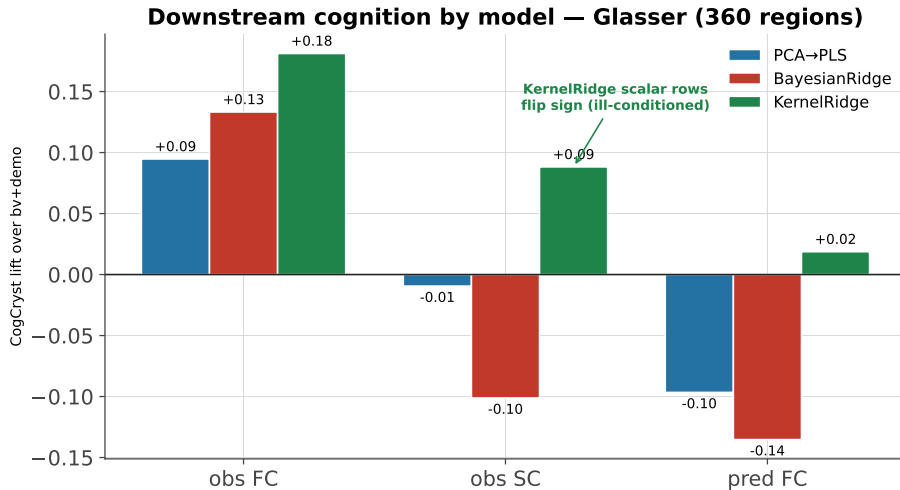
6 · Are the results model-dependent?

Headlines replicate across models; the only divergences are PLS's low oracle and KernelRidge's unstable scalar rows.

reconstruction	PCA→PLS	BayesianRidge	KernelRidge
FC→SC	0.136	0.166	0.133
SC→FC	0.084	0.104	0.081
asymmetry ratio	1.61×	1.60×	1.65×
SC→SC oracle	0.376	0.648	0.648

CogCryst lift	PCA→PLS	BayesianRidge	KernelRidge
obs FC	+0.095	+0.133	+0.181
obs SC	-0.009	-0.101	+0.088
pred FC	-0.097	-0.135	+0.019

6 · Are the results model-dependent?



Takeaways

1. Cross-modal prediction is **real and directional**: $FC \rightarrow SC > SC \rightarrow FC$ ($1.61\times$).
2. But a **cheap bv+demo baseline beats it** at reconstruction — only the *fingerprint/identity* signal is uniquely connectomic.
3. Downstream, signal splits cleanly: **observed FC** carries (crystallized) cognition; **predicted SC** carries family/heritable signal.
4. **Always beat the cheap baseline** before claiming a connectome is useful.

Thank you

Questions?

Backup: deeper dives (Q1, Q2), then 4S456 replication.

Q1 · Can a predicted connectome beat the real one? [BayesianRidge]

Predicted SC beats *observed* SC for cognition — because predicted SC is a projection of FC (the strong modality). It still loses to FC itself.

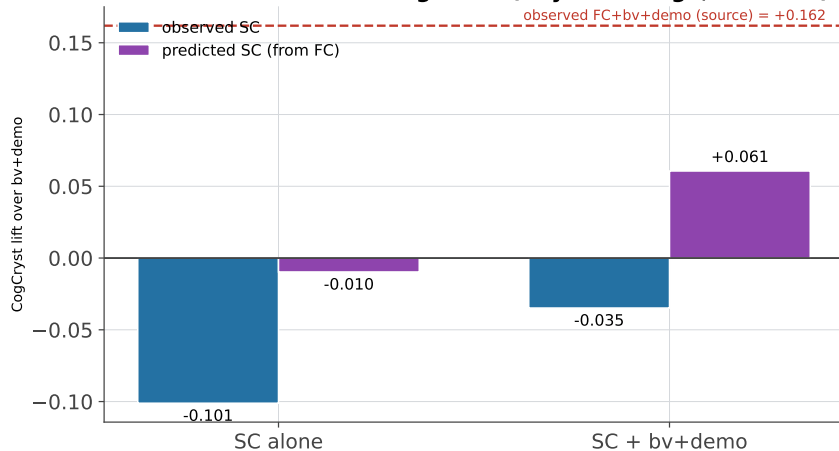
input	Glasser	4S456
observed SC	-0.101	-0.095
predicted SC	-0.010	-0.012
observed SC + bv+demo	-0.035	-0.042
predicted SC + bv+demo	+0.061	+0.056
<i>(source) observed FC + bv+demo</i>	<i>+0.162</i>	<i>+0.146</i>

... but the flip is estimator-specific:

CogCryst lift (Glasser)	PCA→PLS	BayesianRidge	KernelRidge
predicted SC + bv+demo	+0.019	+0.061	+0.017
observed SC + bv+demo	+0.050	-0.035	+0.128
predicted > observed?	no	yes	no

Q1 · Predicted SC beats observed SC — but never the source [BayesianRidge]

Predicted SC beats observed SC — cognition (BayesianRidge, Glasser (360 regions))

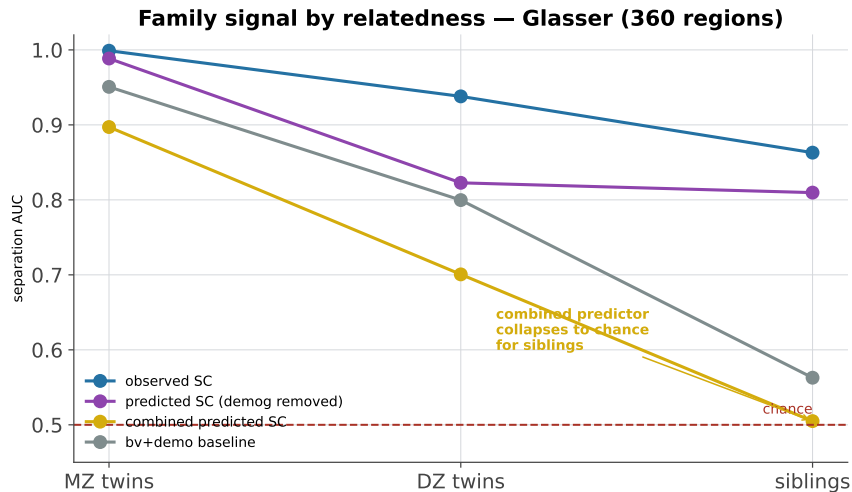


Q2 · Family signal across relatedness (MZ / DZ / sibling) [PCA→PLS]

A genetic dose-response; the reconstruction-optimized *combined* predictor collapses to chance for siblings.

predictor	MZ twins	DZ twins	siblings
observed SC	0.999	0.938	0.863
predicted SC (demog removed)	0.989	0.823	0.810
predicted FC (demog removed)	0.918	0.741	0.743
combined predicted SC	0.897	0.701	0.505
bv+demo baseline	0.951	0.800	0.563

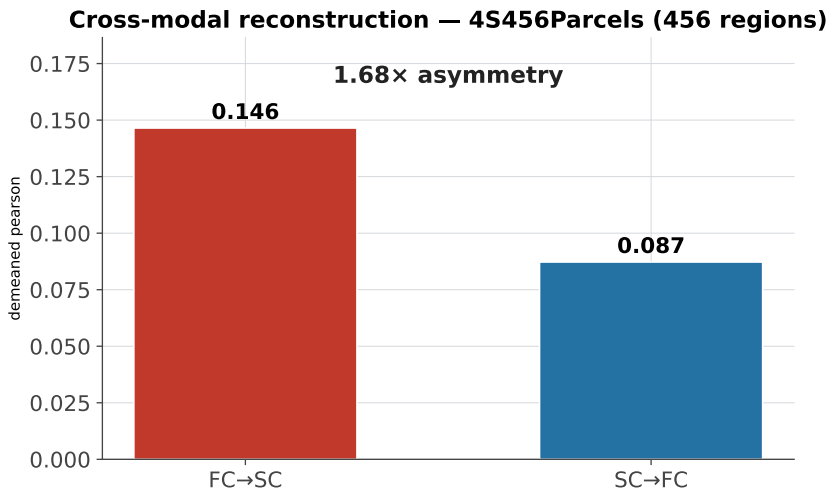
Q2 · Heritability gradient & the objective tradeoff [PCA→PLS]



1 · Cross-modal prediction is directional_[PCA→PLS] (4S456 backup)

direction	demeaned r
FC→SC	0.146
SC→FC	0.087
ratio	1.68×

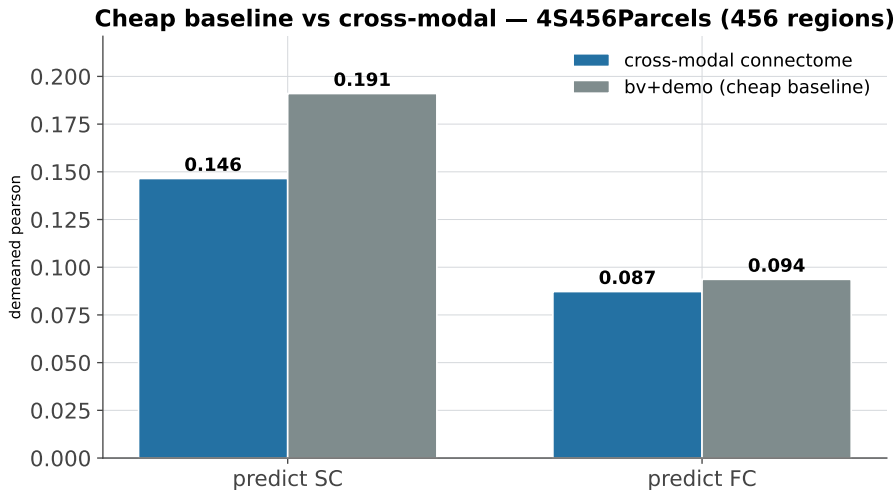
1 · Cross-modal prediction is directional [PCA→PLS] (4S456 backup)



2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS] (4S456 backup)

target	cross-modal	bv+demo
→ SC	0.146	0.191
→ FC	0.087	0.094

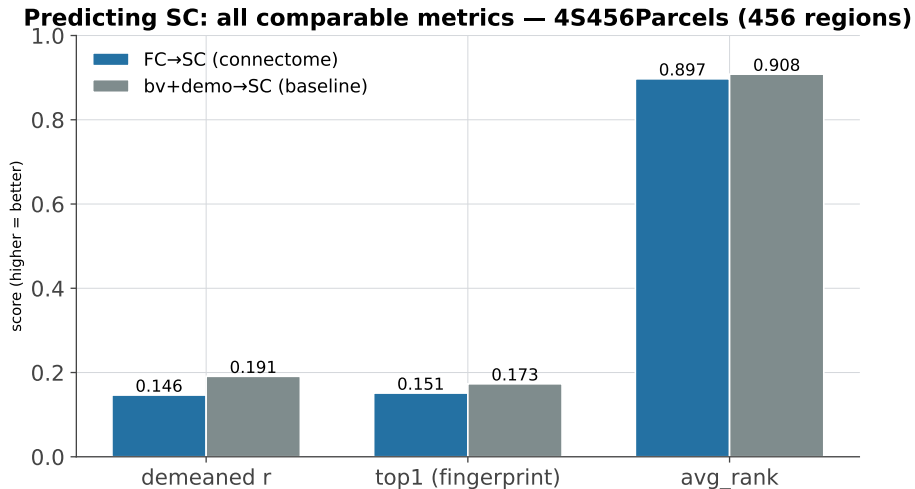
2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS] (4S456 backup)



3a · Predicting SC: every metric [PCA→PLS] (4S456 backup)

input	demeaned r	raw r	top1	avg rank	mse	r²
FC→SC	0.146	0.911	0.151	0.897	0.004	-0.017
bv+demo→SC	0.191	0.913	0.173	0.908	0.004	0.010

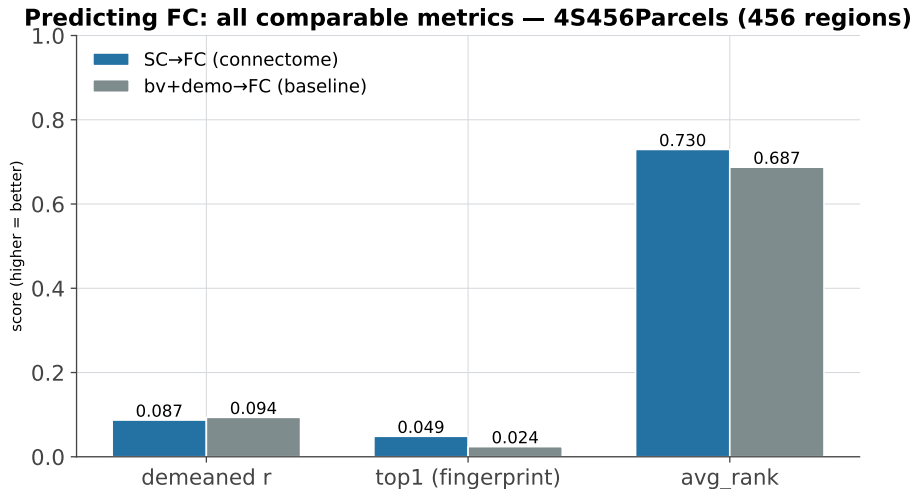
3a · Predicting SC: comparable metrics [PCA→PLS] (4S456 backup)



3b · Predicting FC: every metric [PCA→PLS] (4S456 backup)

input	demeaned r	raw r	top1	avg rank	mse	r^2
SC→FC	0.087	0.813	0.049	0.730	0.013	-0.047
bv+demo→FC	0.094	0.818	0.024	0.687	0.012	-0.020

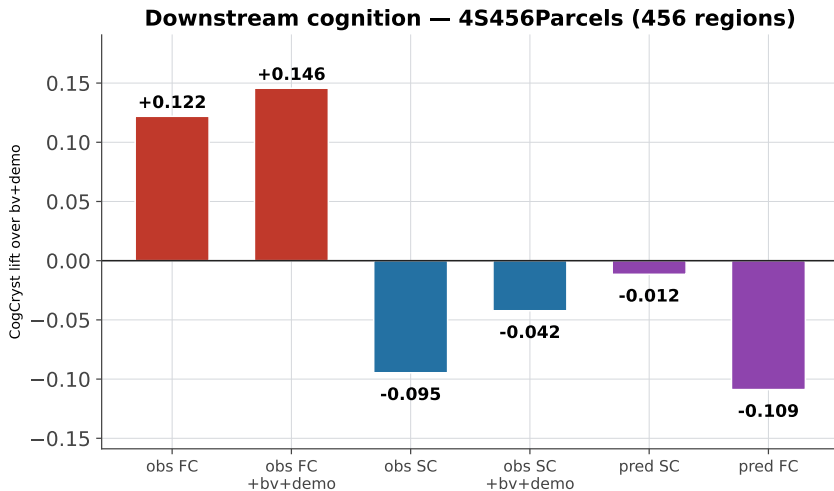
3b · Predicting FC: comparable metrics [PCA→PLS] (4S456 backup)



4 · Downstream cognition: only observed FC helps [BayesianRidge] (4S456 backup)

input	CogCryst r	lift vs bv+demo
observed FC	0.476	+0.122
observed FC + bv+demo	0.500	+0.146
observed SC	0.259	-0.095
observed SC + bv+demo	0.312	-0.042
predicted SC	0.342	-0.012
predicted FC	0.245	-0.109

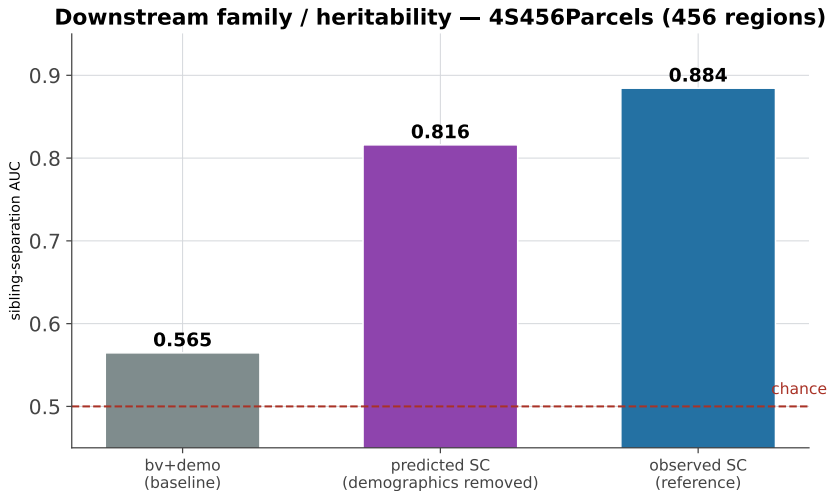
4 · Downstream cognition: only observed FC helps [BayesianRidge] (4S456 backup)



5 · Downstream family: the predicted connectome carries it [PCA→PLS] (4S456 backup)

predictor	sibling AUC
bv+demo (baseline)	0.565
predicted SC (demographics removed)	0.816
observed SC (reference)	0.884

5 · Downstream family: the predicted connectome carries it [PCA→PLS] (4S456 backup)



6 · Are the results model-dependent? (4S456 backup)

Headlines replicate across models; the only divergences are PLS's low oracle and KernelRidge's unstable scalar rows.

reconstruction	PCA→PLS	BayesianRidge	KernelRidge
FC→SC	0.146	0.171	0.143
SC→FC	0.087	0.101	0.083
asymmetry ratio	1.68×	1.70×	1.73×
SC→SC oracle	0.386	0.622	0.621

CogCryst lift	PCA→PLS	BayesianRidge	KernelRidge
obs FC	+0.137	+0.122	+0.176
obs SC	-0.047	-0.095	+0.094
pred FC	-0.101	-0.109	-0.014

6 · Are the results model-dependent? (4S456 backup)

